PROJECT REPORT

ON

"IoT-Based Smart Street Lighting System"

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Under the Guidance of

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COURSE NAME : Internet of Things

COURSE CODE:XCSHA5

Academic Year:2025-2026

CERTIFICATE

This is to certify that the mini-project titled - "**IoT-Based Smart Lighting System"** has been submitted to the Department of CSE, Periyar maniammai institute of science and technology, is a Bonofide record of original work carried out during the academic year 2025 2026 under my supervision and guidance. The project work embodies the results of the student's own efforts and the findings presented in this report have not been submitted to any other University or Institution for the award of any degree or diploma. This certificate is issued as per the requirement of the curriculum prescribed by Periyar Maniammai Institute of Science and Technology.

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DECLARATION

We hereby declare that the mini-project titled "**IoT-Based Smart Lighting System"** submitted in partial fulfillment of the requirements for the award of the degree is an authentic record of my original work.This work has been carried out by us under the guidance of Mr.H.Zahir Hussain Assistant professor, Department of CSE, Periyar Maniammai Institute of Science and Technology. We further declare that this project has not been submitted to any other institution or university for the award of any degree, diploma, or other academic recognition. All the information, data, diagrams, and results presented in this report are genuine and based on the study carried out by us. Any material obtained from other sources has been duly acknowledged.

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ACKNOWLEDGEMENT

We would like to express our deepest gratitude to all those who contributed to the successful completion of this mini-project titled “**IoT-Based Smart Lighting System”.** First and foremost, We convey our heartfelt thanks to our project guide Mr.H.Zahir Hussain for their invaluable guidance, constant support, constructive feedback, and encouragement throughout the course of this work. Their expertise, patience, and suggestions have been instrumental in shaping the quality and direction of this project. We would also like to thank all the faculty members of the department for their help, motivation, and consistent encouragement throughout the duration of our study and project development. Our special thanks go to our classmates, friends, and peers for their cooperation, encouragement, and helpful discussions during the project. Finally, We are deeply grateful to our family for their unconditional love, support, and motivation, without which this project would not have been possible. Their continuous encouragement has always been a source of strength for us.

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Abstract

This project presents an IoT-based smart street lighting system that aims to improve energy efficiency and automation in urban lighting. The system uses a Light Dependent Resistor (LDR) to detect ambient light intensity and automatically control street lights. When the surrounding light level decreases (during night), the street lights are switched ON, and during daylight conditions, they are turned OFF, thereby reducing unnecessary power consumption.

An ESP32 microcontroller is used to process sensor data and control the lighting system. The system is integrated with IoT platforms such as ThingSpeak to monitor and analyze energy usage in real-time. Additionally, a Blynk mobile application is used to provide manual override control, allowing users to turn the street lights ON or OFF remotely when required.

This smart system ensures efficient energy utilization, reduces manual intervention, and supports remote monitoring and control, making it suitable for modern smart city applications.

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Introduction

The rapid growth of urbanization has increased the demand for efficient energy management systems. Conventional street lighting systems operate manually or remain ON for long durations, leading to unnecessary power consumption. To overcome this issue, an IoT-based smart street lighting system is proposed.

This system uses an ESP32 microcontroller integrated with an LDR (Light Dependent Resistor) to automatically control street lights based on ambient light conditions. Additionally, IoT platforms like Blynk and ThingSpeak are used for remote control and real-time monitoring of energy usage.

The system not only reduces energy wastage but also provides flexibility through manual override and brightness control.

Objective

The primary objective of this project is to design and implement an energy-efficient smart street lighting system using IoT technology.

The specific objectives include:

- To automatically control street lights based on ambient light intensity using LDR
- To minimize power consumption and reduce energy wastage
- To implement brightness control using PWM technique
- To enable remote monitoring and control through IoT platforms
- To provide manual override functionality using mobile application
- To analyze energy usage data using cloud platforms
- To develop a cost-effective and scalable solution for smart cities

Literature Survey

Traditional street lighting systems are mostly based on manual switching or timer-based operation. These methods are inefficient as they do not adapt to real-time environmental conditions, resulting in unnecessary power consumption.

Recent research focuses on smart lighting systems that utilize sensors and IoT technologies. Many systems use LDR sensors to detect light intensity and automatically control lighting. Some advanced systems also incorporate motion sensors to further optimize energy usage.

IoT platforms such as ThingSpeak allow real-time monitoring and data analysis, while applications like Blynk provide remote access and control.

Compared to earlier approaches, IoT-based systems offer several advantages:

- Real-time monitoring and data logging
- Remote accessibility and control
- Improved energy efficiency
- Reduced human intervention

This project enhances existing systems by combining automatic control, manual override, and cloud-based monitoring into a single integrated solution.

System Architecture

The proposed system is divided into multiple layers to ensure efficient operation:

1. Sensing Layer

The LDR sensor continuously monitors ambient light intensity and sends analog signals to the controller.

2. Processing Layer

The ESP32 microcontroller processes the input from the LDR and determines the appropriate action (ON/OFF or brightness level).

3. Output Layer

LEDs are used to represent street lights. Their brightness is controlled using PWM signals generated by the ESP32.

4. Communication Layer

The ESP32 connects to WiFi and communicates with IoT platforms.

5. Cloud Layer

- ThingSpeak stores and analyzes data
- Blynk allows user interaction and manual control

OverallFlow:
LDR → ESP32 → LED Control
↘ WiFi → Blynk + ThingSpeak

Components Description

ESP32 Microcontroller

The ESP32 is the core component of the system. It processes sensor data, controls LED brightness, and enables WiFi communication for IoT functionality.

LDR (Light Dependent Resistor)

The LDR senses light intensity. Its resistance changes based on light:

- High light → Low resistance
- Low light → High resistance

This variation is converted into analog values and read by the ESP32.

LEDs (Street Lights)

LEDs act as street lights in this system. Their brightness is adjusted using PWM signals.

Resistors

Resistors are used to limit current and protect components from damage.

WiFi Module

Built into ESP32, it connects the system to the internet for cloud communication.

Blynk Application

Blynk is used to manually control the street lights and adjust brightness using a mobile interface.

ThingSpeak Platform

ThingSpeak is used to store and visualize data such as LDR readings and brightness levels.

Working Principle

The proposed IoT-based smart street lighting system operates in two modes: automatic mode and manual mode, ensuring both efficiency and user control.

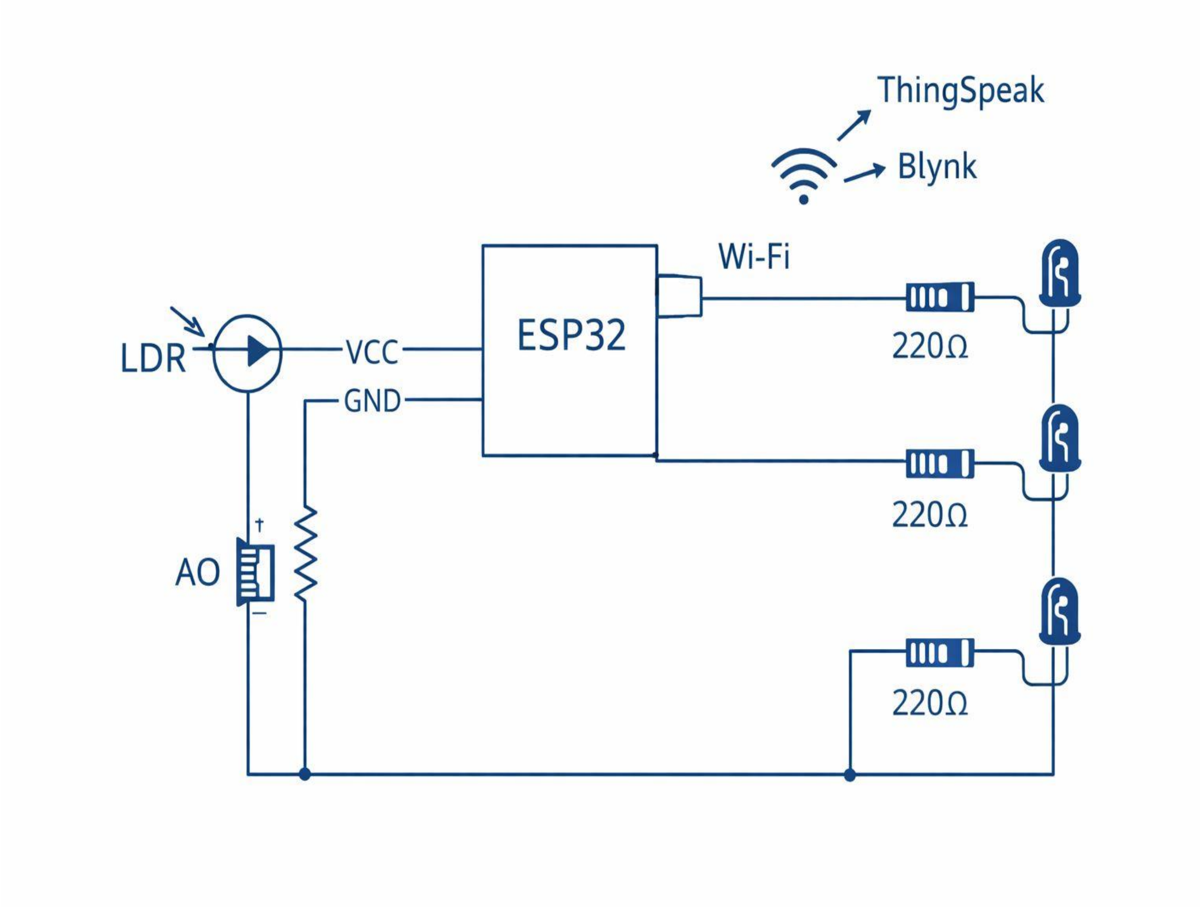
In **automatic mode**, the system uses an LDR sensor to detect ambient light intensity. The sensor continuously sends analog values to the ESP32 microcontroller, which processes the data and controls the street lights accordingly. When the light intensity is high during daytime, the street lights are turned OFF to save energy. As the light level decreases during evening, the lights glow with low brightness, and at night, they operate at full brightness. Based on the programmed logic, if the LDR value is greater than 3000, the lights remain OFF; between 2000 and 3000, the lights glow dimly; between 1000 and 2000, medium brightness is maintained; and below 1000, full brightness is achieved. This automatic adjustment ensures efficient energy utilization and reduces unnecessary power consumption.

In **manual mode**, the user can override the automatic operation using the Blynk application. Through this app, the user can manually switch the street lights ON or OFF and adjust the brightness level using a slider. This feature is especially useful during maintenance or emergency situations.

The brightness of the LEDs is controlled using **PWM (Pulse Width Modulation)** technique. The ESP32 generates PWM signals, and the `ledcWrite()` function is used to vary the intensity of the lights smoothly.

Additionally, the system supports real-time data monitoring using ThingSpeak. The LDR values and brightness levels are sent to the cloud every 15 seconds, enabling performance analysis and energy consumption tracking.

Circuit Diagram



Algorithm

1. **Start**
2. **Initialize System**
   - Initialize serial communication
   - Configure PWM for LED brightness control
   - Connect ESP32 to WiFi network
   - Initialize Blynk platform
   - Initialize ThingSpeak
3. **Read Inputs**
   - Read LDR sensor value (analog input)
   - Receive user inputs from Blynk:
     - Manual mode (V0)
     - ON/OFF control (V1)
     - Brightness level (V2)
4. **Check Mode**
   - If **manual Mode = 0**, go to Automatic Mode
   - Else, go to Manual Mode
5. **Automatic Mode**
   - If LDR value > 3000 → Set brightness = 0 (Lights OFF)
   - Else if LDR value > 2000 → Set brightness = low (dim)
   - Else if LDR value > 1000 → Set brightness = medium
   - Else → Set brightness = high (full brightness)
6. **Manual Mode**
   - If manualState = 1 (ON)
     - Set brightness = value from slider
   - Else
     - Set brightness = 0 (OFF)
7. **Apply Output**
   - Send brightness value to LED using PWM
8. **Send Data to Cloud**
   - Send LDR value to ThingSpeak (Field 1)
   - Send brightness value to ThingSpeak (Field 2)
9. **Wait**
   - Delay for 15 seconds
10. **Repeat Loop**
    - Go back to Step 3
11. **Stop**

Code and Output

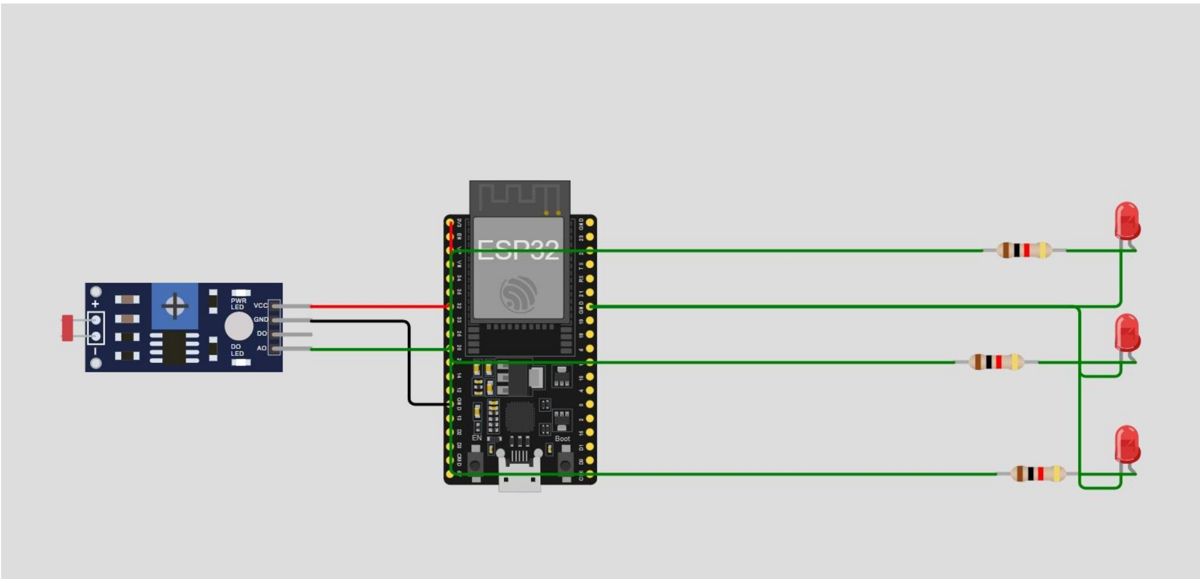
```
#define BLYNK_PRINT Serial

#include <WiFi.h>
#include <BlynkSimpleEsp32.h>
#include <ThingSpeak.h>

// WiFi Credentials
char ssid[] = "YOUR_WIFI_NAME";
char pass[] = "YOUR_WIFI_PASSWORD";

// Blynk Auth Token
char auth[] = "4AyB17v36qX4V1DbPEBpU15xAgVTUX0o";

// ThingSpeak Details
unsigned long channelID = 3330664;
const char * writeAPIKey = "I0RE01BR7XM059T6";

WiFiClient client;

//  Pin Definitions
#define LDR 34        // Analog input
#define LED_PIN 26    // PWM output (all 3 LEDs connected here)

// Variables
int ldrValue = 0;
int manualMode = 0;     // V0 (0 = Auto, 1 = Manual)
```

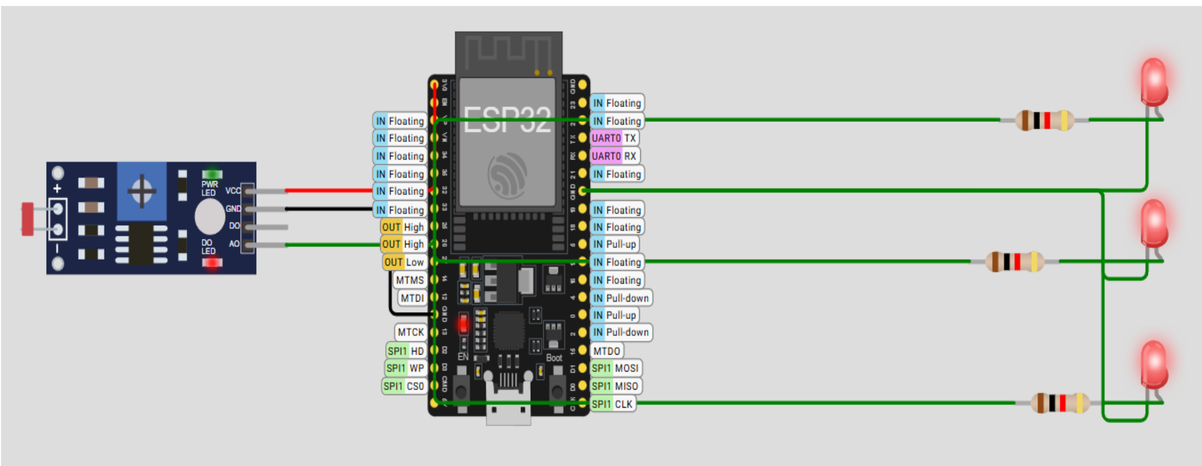
```
int manualState = 0;    // V1 (ON/OFF)
int brightness = 255;   // V2 (optional slider)

//  PWM Setup
const int pwmChannel = 0;
const int pwmFreq = 5000;
const int pwmResolution = 8;

//  Blynk Controls
BLYNK_WRITE(V0) {
  manualMode = param.asInt();
}

BLYNK_WRITE(V1) {
  manualState = param.asInt();
}

BLYNK_WRITE(V2) {   // Optional slider
  brightness = param.asInt();
}

void setup() {
  Serial.begin(9600);

  // PWM setup
  ledcSetup(pwmChannel, pwmFreq, pwmResolution);
  ledcAttachPin(LED_PIN, pwmChannel);
```

```
  // WiFi connect
  WiFi.begin(ssid, pass);
  while (WiFi.status() != WL_CONNECTED) {
    delay(1000);
    Serial.println("Connecting to WiFi...");
  }
  Serial.println("WiFi Connected");

  // Blynk start
  Blynk.begin(auth, ssid, pass);

  // ThingSpeak start
  ThingSpeak.begin(client);
}

void loop() {
  Blynk.run();

  // Read LDR
  ldrValue = analogRead(LDR);
  Serial.print("LDR Value: ");
  Serial.println(ldrValue);

  int outputBrightness = 0;

  // AUTO MODE (Brightness Control)
```

```
if (manualMode == 0) {

  if (ldrValue > 3000) {
    outputBrightness = 0;        // Day → OFF
  }
  else if (ldrValue > 2000) {
    outputBrightness = 80;       // Evening → Dim
  }
  else if (ldrValue > 1000) {
    outputBrightness = 150;      // Medium
  }
  else {
    outputBrightness = 255;      // Night → Full Bright
  }
}

// MANUAL MODE
else {
  if (manualState == 1) {
    outputBrightness = brightness;  // From slider or full
  } else {
    outputBrightness = 0;
  }
}

// Apply brightness
ledcWrite(pwmChannel, outputBrightness);
```

```
  // Send data to ThingSpeak
  ThingSpeak.setField(1, ldrValue);
  ThingSpeak.setField(2, outputBrightness);
  ThingSpeak.writeFields(channelID, writeAPIKey);

  delay(15000);  // ThingSpeak delay
}
```





Results and Discussion

The IoT-based smart street lighting system was successfully implemented using an ESP32 microcontroller, LDR sensor, and LED lights.

- The LDR sensor accurately detected ambient light intensity and automatically controlled the street lights.
- During daytime conditions, the lights remained OFF, reducing unnecessary energy consumption.
- In low-light conditions (evening and night), the system adjusted brightness levels dynamically using PWM.
- The manual override feature using the Blynk app worked effectively, allowing users to control lights remotely.
- Real-time data such as LDR values and brightness levels were successfully transmitted and visualized in ThingSpeak.

Discussion:

The system demonstrated efficient energy utilization and reliable performance. The integration of automation with IoT monitoring ensures reduced manual effort and better control over street lighting systems. However, minor delays were observed due to network latency during cloud updates.

Advantages

- Reduces energy consumption by automatic ON/OFF control
- Provides brightness adjustment using PWM
- Enables remote monitoring and control through mobile application
- Minimizes human intervention
- Cost-effective and easy to implement
- Suitable for smart city applications

Disadvantages

- Initial setup cost is higher compared to traditional systems
- Depends on internet connectivity for IoT features
- Sensor accuracy may vary due to environmental conditions
- Delay may occur in data transmission to cloud platforms

Applications

- Smart city street lighting systems
- Highways and road lighting
- Parking areas and campuses
- Industrial lighting systems
- Residential colony street lights

Future Scope

- Integration with motion sensors to detect vehicle or human movement
- Use of solar panels for renewable energy-based operation
- AI-based prediction for energy optimization
- Fault detection and automatic maintenance alerts
- Integration with smart city infrastructure

Conclusion

The IoT-based smart street lighting system provides an efficient and intelligent solution for modern lighting needs. By using LDR sensors and ESP32, the system automatically controls lighting based on environmental conditions, thereby reducing energy wastage.

The addition of IoT platforms enhances the system by enabling real-time monitoring and remote control. Overall, the project proves to be a reliable, scalable, and energy-efficient solution suitable for smart city applications.

References

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3. Blynk IoT Platform Documentation
4. ThingSpeak IoT Analytics Platform
5. Research papers on Smart Street Lighting Systems
6. Online tutorials and IoT project resources